

Hybrid Level-Flow Graph Entry: Repair Attempt and Exact Remaining Gap

1 The intended hybrid

The target route is not a pure Plan 3 single-level route. It is:

move inward along torsion levels $\implies$ discard an $O(\delta_T(\Omega))$ layer $\implies$ enter the small L^∞ radial-graph

and then use the existing small-amplitude radial-graph theorem plus the superlevel transfer.

Let $u = u_\Omega$, $E_t = \{u > t\}$, and normalize $|\Omega| = \pi$ in dimension two. The exact theorem one would like is:

Conjecture 1.1 (Planar $O(\delta_T)$ graph-entry level). *There are computable dimensional constants $\delta_0, \eta_0, C > 0$ such that if $\Omega \subset \mathbb{R}^2$ is open, $|\Omega| = \pi$, and $\delta_T(\Omega) \leq \delta_0$, then there is a regular level $E_{\hat{t}}$ and a center z such that*

$$|\Omega \setminus E_{\hat{t}}| \leq C\delta_T(\Omega),$$

and, after translating by z and rescaling to area π ,

$$E_{\hat{t}}^\# = \{re^{i\theta} : 0 < r < 1 + \varphi(\theta)\}, \quad \|\varphi\|_{L^\infty(S^1)} \leq \eta_0.$$

If Conjecture 1.1 is true, the rest is already present in the repo. Step 3 gives

$$\delta_T(E_{\hat{t}}) \leq (1 - C\delta_T(\Omega))^{-1}\delta_T(\Omega) \leq 2\delta_T(\Omega),$$

the direct radial-graph theorem `H12_ASYMMETRY_DIRECT_PROOF.tex` gives

$$\mathcal{A}(E_{\hat{t}})^2 \leq C\delta_T(E_{\hat{t}}),$$

and Step 5 gives

$$\mathcal{A}(\Omega) \leq \mathcal{A}(E_{\hat{t}}) + 2 \frac{|\Omega \setminus E_{\hat{t}}|}{|\Omega|}.$$

Thus $\mathcal{A}(\Omega)^2 \leq C\delta_T(\Omega)$. This closure uses no BDV theorem.

2 What the repo supplies

The following ingredients are usable.

1. **Good near-boundary level.** The corrected Step 1 `proof-step1.md` works in the volume variable and gives a regular level with

$$|\Omega \setminus E_{\hat{t}}| \leq C_\theta \delta_T(\Omega), \quad D_H(\hat{t}) \leq C\theta,$$

for fixed absolute $\theta > 0$. The same Markov selection can be made for $D_H + D_I$ at the same fixed scale. The output is a small layer and a small weighted Cauchy–Schwarz defect, not a pointwise gradient floor or a graph.

2. **Exact superlevel deficit transfer.** Step 3 is sound: $u - \hat{t}$ is exactly the torsion function of $E_{\hat{t}}$, and the profile computation gives the displayed bound on $\delta_T(E_{\hat{t}})$.
3. **Transfer back.** Step 5 is a purely measure-theoretic triangle inequality for Fraenkel asymmetry and is sound.
4. **Small-amplitude radial graphs.** The direct radial-graph proof is the right final theorem: it requires a genuine one-crossing radial graph and small L^∞ amplitude, but no slope bound.
5. **Plan 2 slicing technology.** The Maggi radial slicing identity and the bad-ray multiplicity absorption in `wave2-A-radial-l2-trace.md` are useful. In a fixed annular bracket, the number of extra crossings on rays is linearly charged by perimeter excess. This is a real tool, but it controls the measure of bad rays; it does not make the bad-ray set empty.
6. **Self-truncated velocity.** The note `WIP_TrimmedVelocityRepair.tex` correctly extracts from D_H that low-gradient portions of a good level are paid for by the $1/|\nabla u|$ weight. This is the right mechanism for active thin outward fingers in dimension two.

The following cannot be used as load-bearing facts.

1. The old unweighted Hessian Chain B is false. The high-frequency near-ball computation in `PLAN3_N2_SINGLE_LEVEL_RECHECK.tex` shows that no dimensional inequality

$$\delta_T(D) \geq c|D|^{-1} \int_D |D^2 u_D + I/2|^2$$

can hold on all smooth near-balls.

2. The Plan 2 endpoint metric trace is not a literal graph-entry theorem. Even where it gives $d_{\mathcal{X}}(E_\rho, B_\rho)^2 \leq C\delta_T$, this is L^1 -modulo-translation control, not a one-crossing L^∞ radial parametrization.
3. The WIP same-centre FJ package is useful for centering and normal oscillation, but it is not a proof of Conjecture 1.1. It still tolerates small bad-ray sets and imports isoperimetric near-spherical/selection facts, not torsion graph entry.

3 A better replacement for literal graph entry

The correct repair target is slightly weaker than Conjecture 1.1 and appears more realistic.

Conjecture 3.1 (Planar $O(\delta_T)$ radialization lemma). *Let $E = E_{\hat{t}}$ be the Step-1 regular level in dimension two. There exists a set G , after a translation and a rescaling, such that*

$$G = \{re^{i\theta} : 0 < r < 1 + \varphi(\theta)\}, \quad \|\varphi\|_{L^\infty} \leq \eta_0,$$

and

$$|E\Delta G| \leq C\delta_T(\Omega), \quad \delta_T(G) \leq C\delta_T(\Omega). \tag{R}$$

This is enough. The graph theorem gives $\mathcal{A}(G)^2 \leq C\delta_T(G) \leq C\delta_T(\Omega)$, while

$$\mathcal{A}(E) \leq \mathcal{A}(G) + C|E\Delta G|$$

and Step 5 transfers from E to Ω . Notice the important point: the repair volume in (R) must be $O(\delta_T)$, not merely $O(\sqrt{\delta_T})$, unless one has a separate monotonicity argument showing that the repair does not increase the deficit beyond $C\delta_T$.

Conjecture 3.1 is the most promising form of the missing piece because it allows filling tiny holes or deleting tiny loops. Literal graph entry for the actual level is too rigid: a tiny central hole already destroys one-crossing radially on every ray, even though its volume may be much smaller than the torsion deficit.

4 Attempt to prove the radialization lemma

The natural decomposition is along rays from a same center z , using Plan 2's slicing notation. For a.e. angle θ , write

$$E_\theta = \{r > 0 : z + re^{i\theta} \in E\}, \quad \partial^* E_\theta = \{r_{\theta,1} < \cdots < r_{\theta,N(\theta)}\}.$$

Good rays have $N(\theta) = 1$; bad rays have $N(\theta) \geq 2$.

4.1 Good rays

On good rays the radial graph candidate is clear: keep the first crossing $r_{\theta,1}$. The Plan 2 direct T_1 slicing bound controls the angular L^1 radial mismatch by the symmetric difference and perimeter excess. This is useful for metric/Fraenkel closure, but it is not L^∞ control. To obtain small amplitude one still needs an annular containment theorem:

$$B_{1-\eta}(z) \subset E \subset B_{1+\eta}(z), \quad \eta \ll 1. \quad (\text{A})$$

For the Step-1 level, (A) is not proved in the repo. FMP gives measure closeness at the right square-root rate when a pointwise isoperimetric deficit is known; it does not give Hausdorff annular containment for arbitrary finite-perimeter levels. In dimension two, one expects fixed-depth inward slits to cost torsion through capacity or perimeter, but that statement is not currently proved.

4.2 Bad rays

The Plan 2 bad-ray estimate is genuinely helpful:

$$\int_{S^1} (N(\theta) - 1) d\theta \leq C (P(E) - P(B))$$

inside an annular bracket. Thus multiplicity is small in angular measure when the isoperimetric deficit of the level is small.

This still falls short of Conjecture 3.1. First, the Step-1 $O(\delta_T)$ -layer selection only gives D_I small at a fixed absolute threshold if one keeps the layer $O(\delta_T)$. It does not give $D_I(E_i) = O(\delta_T)$. Second, even an $O(\delta_T)$ angular measure bound would not by itself give a graph; it would only say that the non-graph rays form a small set. One must also repair those rays and prove both parts of (R): small symmetric-difference cost and no loss in the torsion deficit.

4.3 Outward fingers

Here dimension two is favorable. On a thin outward tube of width w and length L , active tube levels have

$$D_H(t) \simeq \frac{L}{w}.$$

Thus a Step-1 level with $D_H \leq \theta$ cannot carry a long thin active finger unless $L \lesssim \theta w$. In particular, the volume of such a finger is $wL \lesssim \theta w^2$, and the extra layer needed to clear the tube is of the same w^2 order. This is exactly where $n = 2$ differs from higher dimensions.

This model computation is strong evidence that outward fingers can be radialized at $O(\delta_T)$ cost, but the repo does not contain the general geometric lemma turning this model into a proof for arbitrary bad components of an analytic torsion level.

4.4 Inward fjords and holes

Moving inward cannot fill a missing fjord or a hole. Therefore a literal superlevel graph-entry theorem must prove that no fixed-depth inward defect remains. The realistic statement is instead that the defect can be filled at $O(\delta_T)$ cost while preserving $\delta_T(G) \leq C\delta_T(\Omega)$.

The expected two-dimensional tool is capacity/modulus. A slit or fjord of fixed radial depth should force a definite torsion loss, uniformly in its width, while a small hole of radius r has torsion cost of order $1/|\log r|$, much larger than its area r^2 . This would imply that the area to be filled is $O(\delta_T)$, often much smaller.

But this is still a theorem to prove. The current repo only formulates this capacity idea; it does not provide the needed quantitative decomposition for arbitrary same-ray cancellations.

5 Outcome

The remaining puzzle piece has not been proved by the existing notes. The repo does contain the right components, but they stop one step short of the needed theorem.

The cleanest route forward is:

1. Prove a planar annular-entry/capacity lemma: every fixed-depth inward fjord or hole in a near-optimal torsion level costs a computable positive amount of δ_T ; quantitatively, the total volume that must be filled to obtain annular containment is $O(\delta_T)$.
2. Prove a planar bad-ray radialization lemma in a thin annulus: using Maggi slicing and the Plan 2 multiplicity absorption, the total volume of same-ray multi-crossing/cancellation that must be modified is $O(\delta_T)$, and the modification satisfies $\delta_T(G) \leq C\delta_T(E)$.
3. Feed the resulting radial graph G into `H12_ASYMMETRY_DIRECT_PROOF.tex` and transfer back.

This is still promising, but it is not closed. The underclaim in the older notes was thinking that annular or metric L^1 control is too weak to be useful; it is useful if one allows $O(\delta_T)$ radialization. The overclaim would be saying that the repo already proves this radialization. It does not. The missing theorem is exactly the capacity-plus-bad-ray radialization lemma above.